Homework 3, DDL: May 11, 2026

Python Programming

1. Lab # 5 (4 Pts)
2. Lab # 6 (4 Pts)

Questions for Tech Report (6 Pts, $\leq$ 3 Pages)

1. **Design An Algorithm of "Robust NMF".** When assuming the data noise obeys to Laplace distribution, we can extend classic NMF to "Robust NMF", i.e., given $\boldsymbol{X}_{noisy} \in [0, \infty)^{N\times D}$, solving the following problem:

$$\begin{aligned} &\min\nolimits_{\boldsymbol{U},\boldsymbol{V}} \|\boldsymbol{X}_{noisy} - \boldsymbol{U}\boldsymbol{V}^T\|_1, \\ &s.t.\ \boldsymbol{U} \in [0, \infty)^{N\times L},\ \boldsymbol{V} \in [0, \infty)^{D\times L}. \end{aligned} \tag{23}$$

   Leveraging the algorithms we learned before, can you design an algorithm for robust NMF? (**Hint:** IRLS or RPCA may help you.) (4 Pts)

2. Demonstrate that ISOMAP, LLE, Kernel PCA, and Eigenmap lead to the almost same optimization problem: $\min_{\boldsymbol{Z}\in\mathbb{R}^{N\times L}} \mathrm{tr}(\boldsymbol{Z}^T\boldsymbol{\Phi}\boldsymbol{Z}),\ s.t.\ \boldsymbol{Z} \in \Omega$, and derive the $\boldsymbol{\Phi}$'s and the feasible domain Ω's for the methods. (2 Pts)